

EE 605: Error Correction Codes

Problem Set 4

1. (*List decoding: Singleton bound*) Recall the definition of a (τ, L) list-decodable code from class. A code $\mathcal{C}$ is said to be (τ, L) list-decodable if every ball of radius τn in Q^n contains at most L codewords of $\mathcal{C}$, i.e., for any $y \in Q^n$ we have

$$|B_{\tau n}(y) \cap \mathcal{C}| \leq L$$

where $B_{\tau n}(y)$ is the ball of radius τn centered at y .

Show that for any (τ, L) list-decodable code $\mathcal{C} \subset Q^n$ over an alphabet Q of size q , we have

$$|\mathcal{C}| \leq Lq^{n - \lfloor (L+1)\tau n/L \rfloor}.$$

Note that this generalises the traditional Singleton bound, which corresponds to $L = 1$.

Hint: Let z denote $\lfloor (L+1)\tau n/L \rfloor$. It's ok to assume here that z is an integer and drop the floors. Proceed in the following steps (along similar lines as proof of standard Singleton bound):

- (a) Assume to the contrary that we have a (τ, L) list-decodable code $\mathcal{C}$ with $|\mathcal{C}| > Lq^{n-z}$. The goal is to show that there exists a vector $y \in Q^n$ such that $|B_{\tau n}(y) \cap \mathcal{C}| \geq L + 1$, where $B_{\tau n}(y)$ is a Hamming ball of radius τn around y .
- (b) Use pigeonhole principle to show that there exist at least $L + 1$ distinct codewords $c_1, \dots, c_{L+1}$ which agree on their first $n - z$ coordinates.
- (c) Create a vector y which has the same first $n - z$ coordinates as above. Partition the remaining z coordinates equally into $L + 1$ parts and match each part to a distinct one of the $L + 1$ codewords in the corresponding coordinates.
- (d) Show that the distance of y from each of the $L + 1$ c_i 's is at most τn . Argue contradiction.

Answer: Assume that we have a (τ, L) list-decodable code $\mathcal{C}$ with $|\mathcal{C}| > Lq^{n-z}$. If we pick the first $n - z$, we have $q^{(n-z)}$ possibilities. If there were at most L codewords agreeing on the first $n - z$ coordinates, then we would have $|\mathcal{C}| \leq Lq^{(n-z)}$. Which is a contradiction to our assumption. This means that we have at least L codewords agreeing on the first $(n - z)$ locations. Let these be $c_0, c_1, \dots, c_L$.

Now consider y having the same first $(n - z)$ coordinates. We have a total of q^z such vectors. Partition them among $c_0, c_1, \dots, c_L$ equally. Assume that one of the c_i has distance greater than τn , this is not possible as y has a distance of at most $n - (n - z) - \frac{z}{L+1} = \frac{Lz}{L+1}$ as we have distributed it such that we have z matching coordinates over $c_0, c_1, \dots, c_L$. This shows the contradiction.

2. Prove the following

(a) Show that for the q -ary entropy function given by

$$H_q(x) = -x \log_q x - (1-x) \log_q(1-x) + x \log_q(q-1), \quad x \in [0, 1-1/q]$$

we have $H_q(x) \geq x$. [3 marks]

(b) Show that for the Zyablov code construction with rate

$$R_Z(\delta) = \max_{\Theta \in [\delta, 1-1/q]} (1 - H_q(\Theta)) \left(1 - \frac{\delta}{\Theta}\right),$$

we have

$$(1 - \sqrt{\delta})^2 - \frac{1}{\log_2 q} \leq R_Z(\delta) \leq (1 - \sqrt{\delta})^2.$$

Thus as q becomes large, the Zyablov code construction rate converges to $(1 - \sqrt{\delta})^2$. [5 marks]

Answer:

(a) Consider $f(x) = H_q(x) - x$, if we can show that this function is always greater than 0, then we have proved the required. Taking it's derivative

$$\begin{aligned} f'(x) &= -\log_q x - \frac{1}{\log_q} + \log_q(1-x) + \frac{1}{\log_q} + \log_q(q-1) - 1 \\ f'(x) &= \log_q \left(\frac{1-x}{x} \frac{q-1}{q} \right) \end{aligned}$$

$f'(x)$ is greater than 0 for $x \in \left(0, \frac{q-1}{2q-1}\right)$ and less than 0 for $x \in \left(\frac{q-1}{2q-1}, \frac{q-1}{q}\right)$. $\lim_{x \rightarrow 0} f(x) = 0$, and strictly increases till $x = \frac{q-1}{2q-1}$, the strictly decreases till $x = \frac{q-1}{q}$. $\lim_{x \rightarrow \frac{q-1}{q}} f(x) > 0$, so $f(x)$ is always greater than 0. Which means that the q -ary entropy function is always greater than x .

(b) $R_Z(\delta) = \max_{\Theta \in [\delta, 1-1/q]} (1 - H_q(\Theta)) \left(1 - \frac{\delta}{\Theta}\right)$

$$\Rightarrow R_Z(\delta) \leq \max_{\Theta \in [\delta, 1-\frac{1}{q}]} (1 - \Theta) \left(1 - \frac{\delta}{\Theta}\right) = 1 + \delta - \Theta - \frac{\delta}{\Theta} \leq (1 - \sqrt{\delta})^2$$

(Using the AM-GM inequality) In part a, we got the maximum value of $H_q(x) - x$ at $x = \frac{q-1}{2q-1}$, one substituting, we get it to be $\log_q 2$, so $H_q(\Theta) \leq \Theta + \log_q 2$, substituting the same below

$$\begin{aligned} R_Z(\delta) &\geq \max_{\Theta \in [\delta, 1-\frac{1}{q}]} (1 - \Theta - \log_q 2) \left(1 - \frac{\delta}{\Theta}\right) \\ &\geq 1 + \delta - \Theta - \frac{\delta}{\Theta} - \log_q 2 \quad (\text{Using } \Theta = \sqrt{\delta}) \\ &\geq (1 - \sqrt{\delta})^2 - \frac{1}{\log_2 q} \end{aligned}$$

Thus, we get the relation

$$(1 - \sqrt{\delta})^2 - \frac{1}{\log_2 q} \leq R_Z(\delta) \leq (1 - \sqrt{\delta})^2$$

3. (*List decoding*) In class, we looked at list decoding from errors. Here we look at the case of erasures. Below are some basic definitions relating to list decoding from erasures. For $y \in \{0, 1\}^n$ and $B \subseteq \{1, 2, \dots, n\}$, define $[y]_B \in \{0, 1\}^{|B|}$ to be the projection of y onto the coordinates in B .

Definition 0.1. A binary code $\mathcal{C}$ of blocklength n is said to be (s, L) -erasure list-decodable if for every $r \in \{0, 1\}^{n-s}$ and every set $B \subseteq \{1, 2, \dots, n\}$ of size $(n - s)$, we have $|\{c \in \mathcal{C} : [c]_B = r\}| \leq L$. In other words, given any received word with at most s erasures, the number of codewords consistent with the received word is at most L .

For a set $T \subseteq \{0, 1\}^n$, define the support of T as $\text{supp}(T) = \{i : \exists (x_1, x_2, \dots, x_n) \in T \text{ such that } x_i \neq 0\}$.

Definition 0.2. The r -th generalized Hamming weight of a linear code $\mathcal{C}$, denoted $d_r(\mathcal{C})$, is defined to be the size of the smallest support of an r -dimensional subcode of $\mathcal{C}$.

- (a) Show that a code with minimum distance d is $(d - 1, 1)$ -erasure list-decodable. [1 mark]
- (b) Show that the minimum distance d of a code $\mathcal{C}$ is equivalent to $d_1(\mathcal{C})$. [1 mark]
- (c) Show that for any linear code $\mathcal{C}$, $\mathcal{C}$ is (e, L) -erasure list-decodable if and only if $d_r(\mathcal{C}) > e$ for $r = 1 + \log_2 L$. [5 marks]

Answer:

- (a) When the list size is 1, erasure-list decoding is the same as the regular erasure decoding. Any two codes will differ with each other on at least d positions, so if we perform $d - 1$ erasures, we should still be able to decode the received codeword. So, a code of minimum distance d is $(d - 1, 1)$ erasure-list decodable.
- (b) A 1-dimensional simply consists of a non-zero codeword and the zero codeword. The smallest support of this subcode is the same as the minimum weight of the non-zero codeword. Since the minimum weight is d , we get that it is equivalent to $d_1(\mathcal{C})$.
- (c) Suppose $\mathcal{C}$ is (e, L) -erasure list-decodable. Let $c_1, c_2, \dots, c_r$ be r linearly independent codewords in $\mathcal{C}$ which span a subcode $\mathcal{C}'$ of dimension r . We wish to show that $|\text{supp}(\mathcal{C}')| > e$. Consider the received word $\mathbf{r}$ that has all positions in $\text{supp}(\mathcal{C}')$ erased and 0's in the remaining positions. Clearly, each of the $2^r > L$ codewords in $\mathcal{C}'$ is consistent with $\mathbf{r}$, which together with the (e, L) -erasure list-decodability of $\mathcal{C}$ implies that $\text{supp}(\mathcal{C}') > e$.

Conversely, suppose $d_r(\mathcal{C}) > e$. Let $\mathbf{r}$ be a received word with e erasures; let T be the set of unerased positions. The set of codewords of $\mathcal{C}$ consistent with $\mathbf{r}$ is an affine space consisting of solutions to the linear system $\mathbf{c}_{|T} = \mathbf{r}_{|T}$ (here, $\mathbf{x}_{|T}$ denotes the projection of $\mathbf{x}$ onto coordinates in T). Let this set be $\{c_1, \dots, c_M\}$; we wish to prove $M \leq L$. Consider the set $D = \{c_i - c_1 : 1 \leq i \leq M\}$. Then D is a linear subspace of $\mathcal{C}$ with $\text{supp}(D) \cap T = \emptyset$, and hence, $|\text{supp}(D)| \leq e$. Using $d_r(\mathcal{C}) > e$, this means $\dim(D) \leq r - 1 = \lfloor \log L \rfloor$, implying

$$M = |D| = 2^{\dim(D)} \leq 2^{\lfloor \log L \rfloor} \leq L.$$

4. (*List decoding from erasures*) We say that a code is (p, L) -erasure list-decodable if for any vector $y \in \{0, 1, ?\}^n$ (where $?$ denotes the erasure symbol) with at most pn erasures, there are at most L codewords that agree with y in the unerased positions. For any vector c and $T \subset [n]$, let c_T denote the restriction of c to T , i.e., it is the $|T|$ -length vector $(c_i : i \in T)$. Formally, a code $\mathcal{C} \subset F_2^n$ is (p, L) -erasure list-decodable if for every $T \subset [n]$ with $|T| \geq (1-p)n$, and $y' \in \{0, 1\}^{|T|}$, we have

$$|\{c \in \mathcal{C} : c_T = y'\}| \leq L.$$

Prove the following:

- (a) For every $\epsilon > 0$, there exists a (p, L) -erasure list decodable code of rate

$$R \geq \frac{L}{L+1}(1-p) - \frac{H(p)}{L+1} - \epsilon$$

Hint: Use random codes, as done in class for list decoding from errors. For a fixed T, y' , compute the probability that a given set of $L+1$ codewords are all equal to y' when restricted to T . Then take a union bound over sets of codewords, y' , and T .

Answer: Consider a random code $C : \{0, 1\}^k \rightarrow \{0, 1\}^n$. Fix T and y' . Consider a subset of the code $S \subset C, |S| = L+1$ and we have $|T| \geq (1-p)n$. We have $P[C(S) = y'] = \left(\frac{1}{2^{|T|}}\right)^{L+1}$, where $P[\cdot]$ is the probability of an event. Let the probability of a random code not being (p, L) -erasure list-decodable be q

$$\begin{aligned} q &\leq \sum_{T \subset [n], |T| \geq (1-p)n} \binom{2^k}{L+1} 2^{|T|} \left(\frac{1}{2^{|T|}}\right)^{L+1} \binom{n}{|T|} \\ q &\leq \binom{2^k}{L+1} \sum_{T \subset [n], |T| \geq (1-p)n} 2^{-L|T|} \\ q &\leq \binom{2^k}{L+1} np 2^{-L|T|} \\ q &\leq p \cdot 2^{n[R(L-1)+1-L(1-p)]} \\ q &\leq p \cdot 2^{n[1-H(p)-\epsilon(L+1)]} \\ q &\leq 2^{n[1-\epsilon(L+1)]} \end{aligned}$$

The above is negligible when in the limit. Thus, the (p, L) erasure list decodable code exists.

5. Graph-based codes

- (a) Draw the Tanner graph of the binary linear code with parity check matrix

$$H = \begin{bmatrix} 1 & 0 & 0 & 0 & 1 & 1 & 0 & 1 \\ 0 & 1 & 0 & 0 & 1 & 0 & 1 & 1 \\ 0 & 0 & 1 & 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 & 0 & 1 & 1 & 0 \end{bmatrix}.$$

- (b) Recall from class that an $(n, m, D, \gamma, \alpha)$ bipartite expander graph is a D -left regular bipartite graph $G(L \cup R, \mathcal{E})$ where $|L| = n$, $|R| = m$ and $\forall S \subseteq L$ with $|S| \leq \gamma n$, $N(S) \geq \alpha|S|$. Show that the graph in the figure above is a $(7, 5, 2, 2/7, 3/2)$ bipartite expander.

